The documentation and process conversion measures necessary to comply with this document shall be completed by 28 October 2016.

INCH-POUND

MIL-PRF-19500/37F
28 July 2016
SUPERSEDING
MIL-PRF-19500/37E
w/AMENDMENT 1
16 December 2011

PERFORMANCE SPECIFICATION SHEET

* TRANSISTOR, NPN, SILICON, LOW-POWER TYPES 2N333, 2N335, 2N336, JAN

Inactive for new design after 7 June 1999

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

* 1.1 Scope. This specification covers the performance requirements for silicon, NPN, low-power transistors. One level of product assurance (JAN) are provided for each device type as specified in MIL-PRF-19500.

* 1.2 Package outlines. The device packages outline for this specification sheet are a TO-5 (for isolated leads only) in accordance with figure 1 and TO-205AA in accordance with figure 2 for all encapsulated device types.

* 1.3 Maximum ratings. Unless otherwise specified, T_A = +25°C.

Types	P_T T_A = 25°C	V_{CEO}	V_{CBO}	V_{EBO}	T_{stg}
	mW	V dc	V dc	V dc	°C
2N333, T2, LT2 2N335, T2, LT2 2N336, T2, LT2	(1) 150	45	45	1	-65 to +175
2N333A, T2, LT2 2N335A, T2, LT2 2N336A, T2, LT2	(2) 500	45	45	4	-65 to +175

(1) Derate approximately 1 mW/°C for T_A between +25°C and +175°C.

(2) Derate approximately 3.33 mW/°C for T_A between +25°C and +175°C.

* Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil . Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A

FSC 5961

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* 1.4 Primary electrical characteristics. Unless otherwise specified, T_A = +25°C.

Limits	h_{fe} V_{CB} = 5 V dc, I_E = -1 mA			f_{hfb} V_{CB} = 5 V dc I_E = -1 mA dc	C_{obo} V_{CB} = 5 V dc I_E = 0 mA dc 100kHz ≤ f ≤ 1 MHz
	2N333, T2, LT2 2N333A, T2, LT2	2N335, T2, LT2 2N335A, T2, LT2	2N336, T2, LT2 2N336A, T2, LT2		
				MHz	pF
Min	18	37	76	2.5	
Max	44	90	270		15

* 1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example, 6.5 for a list of available PINs.

* 1.5.1 JAN certification mark and quality level. The only quality level designator for encapsulated devices that is applicable for this specification sheet is the base quality level "JAN" that uses no modifiers.

* 1.5.2 Device type. The designation system for the device types of transistors covered by this specification sheet are as follows.

* 1.5.2.1 First number and first letter symbols. The transistors of this specification sheet use the first number and letter symbols "2N".

* 1.5.2.2 Second number symbols. The second number symbols for the transistors covered by this specification sheet are as follows: "333, 335, and 336".

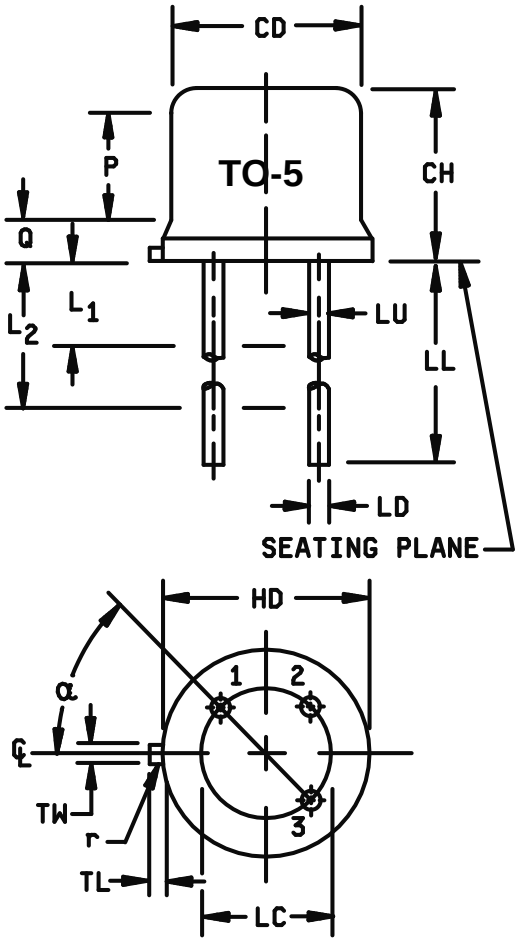
* 1.5.3 Suffix symbols. The following suffix letters are incorporated in the PIN in the order listed in the table as applicable:

	A blank first suffix symbol indicates a 150 mW device.
A	The first suffix symbol "A" indicates a 500 mW device.
	A blank second symbol indicates a TO-5 with all leads electrically isolated from the case.
T2	A “T2” second suffix indicates a TO-205AA with the collector internally connected to the case.
L	An "L" suffix indicates dimension LL is 1.50 (38.10 mm) minimum, 1.75 (19.05 mm) maximum.

* 1.5.4 Lead finish. The lead finishes applicable to this specification sheet are listed on QML-19500.

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CD	.305	.335	7.75	8.51	
CH	.240	.260	6.10	6.60	
HD	.335	.370	8.51	9.40	
LC	.200 TP		5.08 TP		6
LD	.016	.021	0.41	0.53	7, 8
LL	1.500	1.750	38.10	44.45	7, 8
LU	.016	.019	0.41	0.48	7, 8
L_1		.050		1.27	7, 8
L_2	.250		6.35		7, 8
P	.100		2.54		
Q		.050		1.27	5
TL	.029	.045	0.74	1.14	3, 4
TW	.028	.034	0.71	0.86	3
r		.010		0.25	10
α	45°TP		45°TP		6

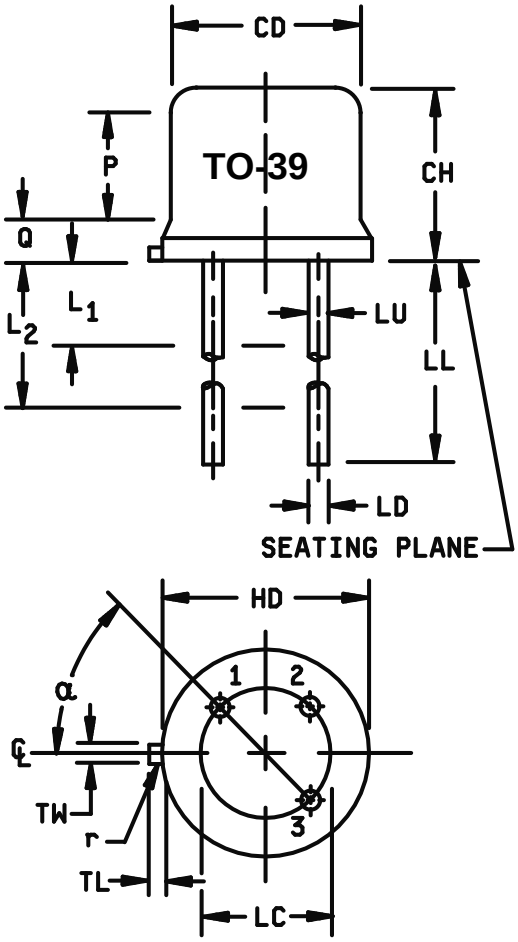


NOTES:
1. Dimension are in inches.
2. Metric equivalents are given for general information only.
3. Beyond r (radius) maximum, TW shall be held for a minimum length of .011 (0.28 mm).
4. Dimension TL measured from maximum HD.
5. Body contour optional within zone defined by HD, CD, and Q.
6. Leads at gauge plane .054 +.001 -.000 inch (1.37 +0.03 -0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC. The device may be measured by direct methods.
7. Dimension LU applies between L_1 and L_2. Dimension LD applies between L_2 and minimum. Diameter is uncontrolled in L_1 and beyond LL minimum.
8. All leads electrically isolated from the case.
9. Dimension r (radius) applies to both inside corners of tab.
10. In accordance with ANSI Y14.5M, diameters are equivalent to Φx symbology.

FIGURE 1. Physical dimensions of transistors, types 2N333, 2N333A, 2N335, 2N335A, 2N336, and 2N336A (TO-5).

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CD	.305	.335	7.75	8.51	
CH	.240	.260	6.10	6.60	
HD	.335	.370	8.51	9.40	
LC	.200 TP		5.08 TP		6
LD	.016	.021	0.41	0.53	7, 8
LL	.500	.750	12.70	19.05	7, 8, 12
LU	.016	.019	0.41	0.48	7, 8
L_1		.050		1.27	7, 8
L_2	.250		6.35		7, 8
P	.100		2.54		
Q		.050		1.27	5
TL	.029	.045	0.74	1.14	3, 4
TW	.028	.034	0.71	0.86	3
r		.010		0.25	10
α	45°TP		45°TP		6



NOTES:

1. Dimension are in inches.
2. Metric equivalents are given for general information only.
3. Beyond r (radius) maximum, TW shall be held for a minimum length of .011 (0.28 mm).
4. Dimension TL measured from maximum HD.
5. Body contour optional within zone defined by HD, CD, and Q.
6. Leads at gauge plane .054 +.001 -.000 inch (1.37 +0.03 -0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC. The device may be measured by direct methods.
7. Dimension LU applies between L_1 and L_2. Dimension LD applies between L_2 and minimum. Diameter is uncontrolled in L_1 and beyond LL minimum.
8. All three leads.
9. The collector shall be internally connected to the case.
10. Dimension r (radius) applies to both inside corners of tab.
11. In accordance with ANSI Y14.5M, diameters are equivalent to Φx symbology.
12. For "L" suffix devices, dimension LL is 1.50 (38.10 mm) minimum, 1.75 (19.05 mm) maximum.

FIGURE 2. Physical dimensions of transistors, types 2N333T2, 2N333AT2, 2N335T2, 2N335AT2, 2N336T2, 2N336AT2 2N333LT2, 2N333ALT2, 2N335LT2, 2N335ALT2, 2N336LT2, and 2N336ALT2 (TO-205AD).

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 – Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 – Test Methods for Semiconductor Devices.

* (Copies of these documents are available online at http://quicksearch.dla.mil.)

* 2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

* 3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and figure 1 (TO-5) and figure 2 (TO-205AA) herein.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

* 3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and tables I and table II.

3.6 Electrical test requirements. The electrical test requirements shall be group A as specified herein.

3.7 Marking. Marking shall be in accordance with MIL-PRF-19500. At the option of the manufacturer, the following marking may be omitted from the body of the transistor.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

* 4.3 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{MD} (and V_C where appropriate).

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table E-V of MIL-PRF-19500 and table I herein.

* 4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with conditions specified for subgroup testing in table E-VIb (JAN) of MIL-PRF-19500 as applicable.

* 4.4.2.1 Group B inspection, (JAN). Separate samples may be used for each step. In the event of a lot failure, the resubmission requirements of MIL-PRF-19500 shall apply. In addition, all catastrophic failures during CI, (conformance inspection), shall be analyzed to the extent possible to identify root cause and corrective action. Whenever a failure is identified as wafer lot or wafer processing related, the entire wafer lot and related devices assembled from the wafer lot shall be rejected unless an appropriate determined corrective action to eliminate the failures mode has been implemented and the devices from the wafer lot are screened to eliminate the failure mode.

Step	Method	Condition
1	1026	Steady-state life: 1,000 hours minimum, V_{CB} = 10 V dc, power shall be applied to achieve T_J = +150°C minimum using a minimum of P_D = 75 percent of maximum rated P_T as defined in 1.3. n = 45 devices, c = 0. The sample size may be increased and the test time decreased as long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours.
2	1048	Blocking life, T_A = +150°C, V_{CB} = 80 percent of rated voltage, 48 hours minimum. n = 45 devices, c = 0.
3	1032	High-temperature life (non-operating), t = 340 hours, T_A = +175°C. n = 22, c = 0.

* 4.4.2.2 Group B sample selection. Samples selected from group B inspection shall meet all of the following requirements:

a. For JAN samples shall be selected randomly from a minimum of three wafers (or from each wafer in the lot) from each wafer lot. See MIL-PRF-19500.

b. Shall be chosen from an inspection lot that has been submitted to and passed table I, subgroup 2, conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (group B for JAN) may be pulled prior to the application of final lead finish.

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* 4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with conditions specified for subgroup testing in table E-VII of MIL-PRF-19500

4.4.3.1 Group C inspection (JAN), table E-VII of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition E.
C6		Not applicable.

4.4.3.2 Group C sample selection. Samples for subgroups in group C shall be chosen at random from any inspection lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes table I tests herein for conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for C6 life test may be pulled prior to the application of final lead finish. Testing of a subgroup using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.

* 4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table II herein.

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* TABLE I. Group A inspection.

	Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
		Method	Details		Min	Max	
	Subgroup 1 2/						
*	Visual and mechanical examination 3/	2071					
*	Solderability 3/	2026	n = 15 leads, c = 0				
*	Resistance to solvents 3/	1022	n = 15 devices, c = 0 (not required for laser marked devices).				
*	Salt atmosphere (corrosion)	1041	n = 6 devices, c = 0 (laser devices only).				
*	Temp cycling 3/	1051	Test condition C, 25 cycles. n = 22 devices, c = 0				
	Subgroup 2						
*	Thermal impedance	3131	See 4.3	$Z_{\theta JX}$			°C/W
	Collector to emitter cutoff current	3041	Bias condition D; V_{CE} = 30 V dc	I_{CEO}		1.0	µA dc
	Collector to base cutoff current	3036	Bias condition D; V_{CB} = 30 V dc	I_{CBO1}		0.5	µA dc
	Breakdown voltage, collector to base	3001	Bias condition D; I_C = 50 µA dc	BV_{CBO}	45		V dc
	Breakdown voltage, collector to emitter	3011	Bias condition D; I_C = 1 mA dc	BV_{CEO1}	45		V dc
	Breakdown voltage, collector to emitter	3011	Bias condition D; I_C = 100 µA dc	BV_{CEO2}	45		V dc
	Breakdown voltage, emitter to base	3026	Bias condition D	BV_{EBO2}			
	2N333, 2N335, 2N336, T2, LT2 2N333A, 2N335A, 2N336A, T2, LT2		I_E = 10 µA dc I_E = 100 µA dc		1.0 4.0		V dc V dc
	Collector to emitter voltage (saturated)		I_C = 5 mA dc	$V_{CE(sat)}$			
	2N333, 2N335, 2N336, T2, LT2 2N333A, 2N335A, 2N336A, T2, LT2		I_B = 2.2 mA dc I_B = 1.0 mA dc			1.0 1.0	V dc V dc
	Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 10 mA dc	h_{FE1}			
	2N333, 2N333A, T2, LT2 2N335 2N335A, T2, LT2 2N336, 2N336A, T2, LT2				6 12 25	60 120 275	

See footnotes at end of table.

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* TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Details		Min	Max	
Subgroup 2 - Continued						
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_E = -1 mA dc	h_{FE2}			
2N333, 2N333A, T2, LT2				8		
2N335 2N335A, T2, LT2				18		
2N336, 2N336A, T2, LT2				35		
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 100 µA dc	h_{FE3}			
2N333, 2N333A, T2, LT2				3		
2N335 2N335A, T2, LT2				6		
2N336, 2N336A, T2, LT2				15		
Subgroup 3						
High-temperature operation:		T_A = +150°C				
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 30 V dc	I_{CBO2}		25	µA dc
Subgroup 4						
Small-signal open-circuit output admittance	3216	V_{CB} = 5 V dc; I_E = -1 mA dc	h_{ob}	0	1.2	µmho
Small-signal open-circuit reverse-voltage transfer ratio	3211	V_{CB} = 5 V dc; I_E = -1 mA dc	h_{rb}	0	$1x10^{-3}$	
Small-signal open-circuit input impedance	3201	V_{CB} = 5 V dc; I_E = -1 mA dc	h_{ib}			
2N333, 2N333A				30	80	ohms
2N335, 2N335A						
2N336, 2N336A						
*For T2, and LT2 only.				20	80	ohms

See footnotes at end of table.

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* TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Details		Min	Max	
Subgroup 4 - Continued						
* Small-signal open-circuit forward-current transfer ratio	3206	V_{CB} = 5 V dc; I_E = -1 mA dc	h_{fe}			
2N333, 2N333A, T2, LT2				18	44	
2N335, 2N335A, T2, LT2				37	90	
2N336, 2N336A, T2, LT2				76	270	
Magnitude of common-emitter small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 5 V dc; I_C = 1 mA dc; f = 2.5 MHz	$\lvert h_{fe1} \rvert$			
2N333, T2, LT2					40	
2N335, T2, LT2					60	
2N336, T2, LT2					120	
2N333A, T2, LT2					30	
2N335A, T2, LT2					30	
2N336A, T2, LT2					30	
Magnitude of common-emitter small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 5 V dc; I_C = 1 mA dc; f = 10 MHz	$\lvert h_{fe2} \rvert$			
2N333, T2, LT2					13	
2N335, T2, LT2					15	
2N336, T2, LT2					17	
2N333A, T2, LT2					13	
2N335A, T2, LT2					13	
2N336A, T2, LT2					13	
Small-signal short-circuit forward-current transfer-ratio, cutoff frequency	3301	V_{CB} = 5 V dc; I_E = -1 mA dc	f_{hfb}	2.5		MHz
Subgroup 5						
Open circuit output capacitance	3236	V_{CB} = 5 V dc; V_{sig} = 0.1 Vac; 100 kHz ≤ f ≤ 1MHz	C_{obo}		15	pF
Noise figure	3246	V_{CB} = 5 V dc; I_E = -1 mA dc; f = 1kHz; R_g = 500 ohms	NF		30	dB

1/ For sampling plan see MIL-PRF-19500.
2/ For resubmission of failed test in subgroup 1 of table I, double the sample size of the failed test or sequence of tests. A failure in table I, subgroup 1 shall not require retest of the entire subgroup. Only the failed test shall be rerun upon submission.
3/ Separate samples may be used.

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* TABLE II. Group E inspection (all quality levels) – for qualification and requalification only.

Inspection	MIL-STD-750		Sample plan
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycling (air to air)	1051	Test condition C, 500 cycles.	
Hermetic seal Fine leak Gross leak	1071		
End-point electrical measurements		See table I, subgroup 2 herein.	
Subgroup 2			45 devices c = 0
Intermittent operating life	1037	V_{CE} = 40 V dc, 6,000 cycles. Adjust device current, or power, to achieve a minimum ΔT_J of 100°C.	
End-point electrical measurements		See table I, subgroup 2 herein.	
Subgroup 4			Sample size N/A
Thermal impedance curves		See MIL-PRF-19500.	
Subgroup 5			
Not applicable			
Subgroup 8			45 devices c = 0
Reverse stability	1033	Condition B.	

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

* 6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

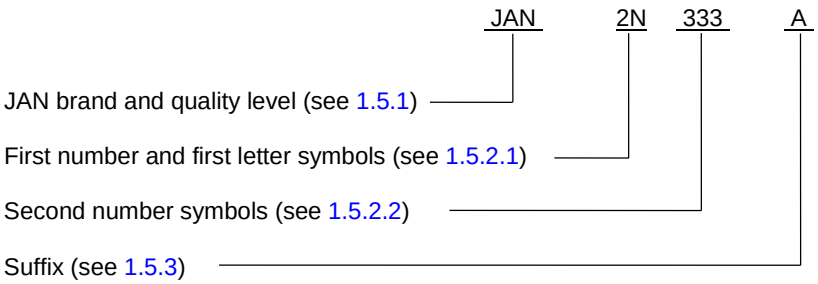
b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

* d. The complete Part or Identifying Number (PIN), see title and section 1.5.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil/.

* 6.4 PIN construction example. The PINs for encapsulated devices are construction using the following form.



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* 6.5 List of PINs. The following is a list of possible PINs available on this specification sheet.

PINs for types 2N333, 2N335, 2N336, 2N333A, 2N335A, and 2N336A		
JAN2N33	JANT2N335	JAN2N336
JAN2N333A	JANT2N335A	JAN2N336A
JAN2N333T2	JAN2N335T2	JAN2N336T2
JAN2N333LT2	JAN2N333LT2	JAN2N336LT2
JAN2N333AT2	JAN2N335AT2	JAN2N336AT2
JAN2N333ALT2	JAN2N335ALT2	JAN2N336ALT2

* 6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Review activities:
Army - AR, MI
Navy - AS, MC
Air Force - 19

Preparing activity:
DLA - CC

(Project 5961-2016-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:00Z
Check the source to verify that this is the current version before use.